

From hyperspectral CHRIS to four RGBN bands

The cloud model eats **4 bands** (R, G, B, NIR). CHRIS has **18 to 62 bands** per mode. So each pseudo-band is built from the CHRIS bands that fall in it.

Three ways to build them: the plain mean the model was trained on, then SRF-weighted resampling, then convolution with the CHRIS response.

METHOD 1 OF 3

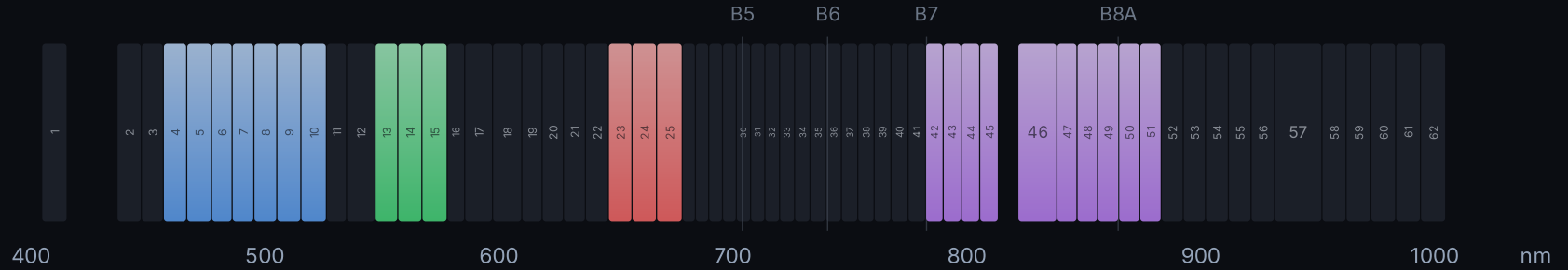
Plain mean

- which each CHRIS band goes to the Sentinel-2 band it overlaps the most
- how plain **mean** of the CHRIS bands assigned to each pseudo-band
- then DN/100 000 or TOA/10 000, clip to 5, and a per-tile z-score inside the network

$$\bar{L}_g = \frac{1}{N_g} \sum_{i \in g} L_i$$

This is the training mapping, the one the model actually learned.

Mode 1 : 62 bands, ~34 m



B2
459-525 nm



B3
542-578 nm

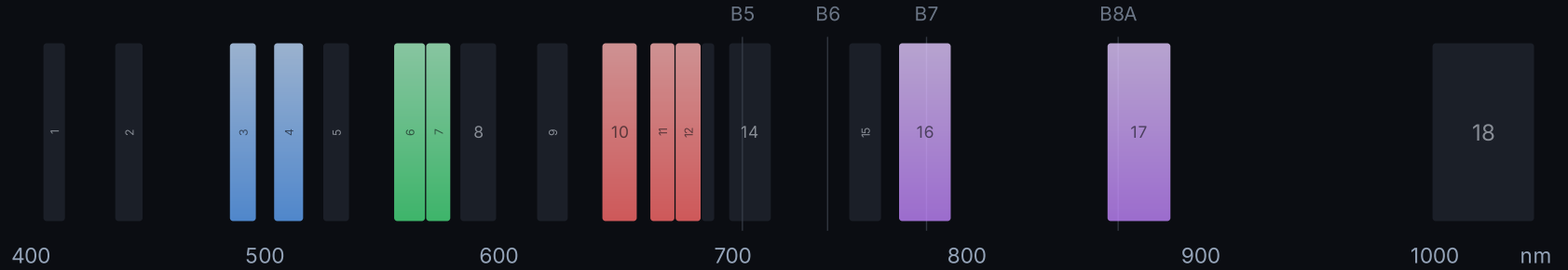


B4
649-680 nm

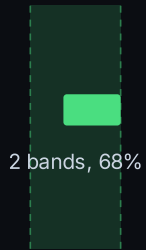


B8
780-886 nm

Mode 2 : 18 bands, ~17 m



B2
459-525 nm



B3
542-578 nm



B4
649-680 nm



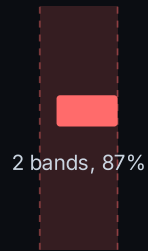
B8
780-886 nm

Mode 3 : 18 bands, ~13 m

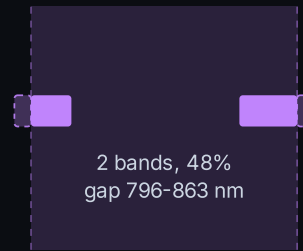


B2
459-525 nm

B3
542-578 nm

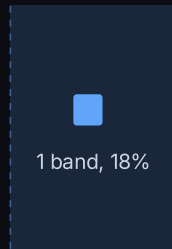


B4
649-680 nm

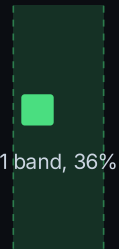


B8
780-886 nm

Mode 4 : 18 bands, ~11 m



B2
459-525 nm



B3
542-578 nm

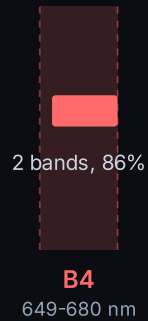
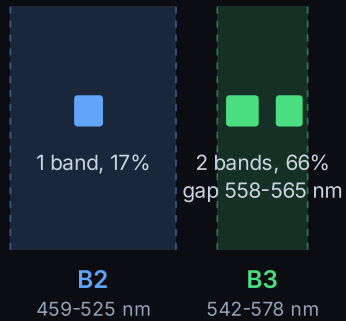
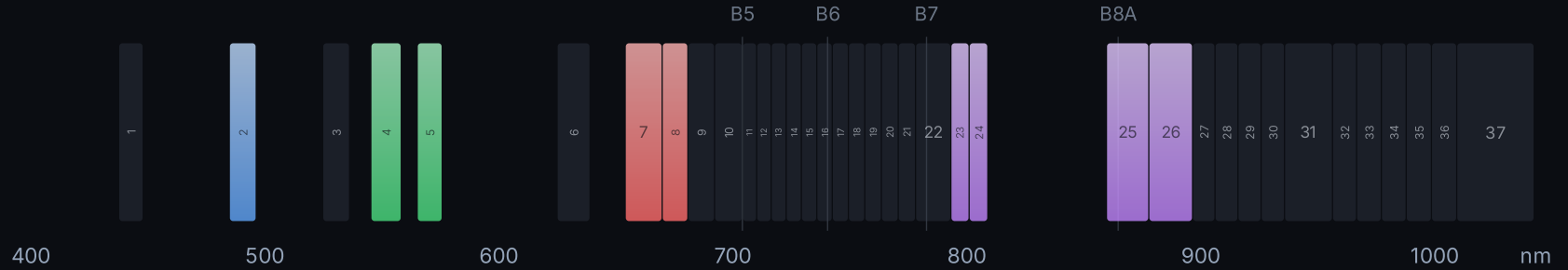


B4
649-680 nm

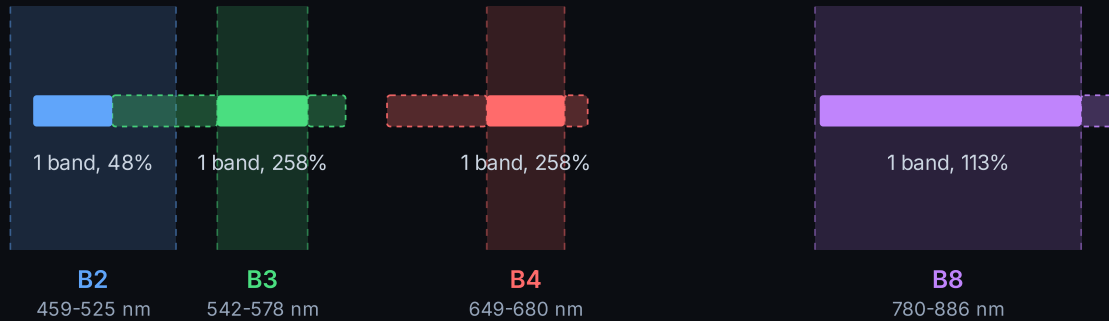


B8
780-886 nm

Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Method 1: where the bands end up

R, B4

G, B3

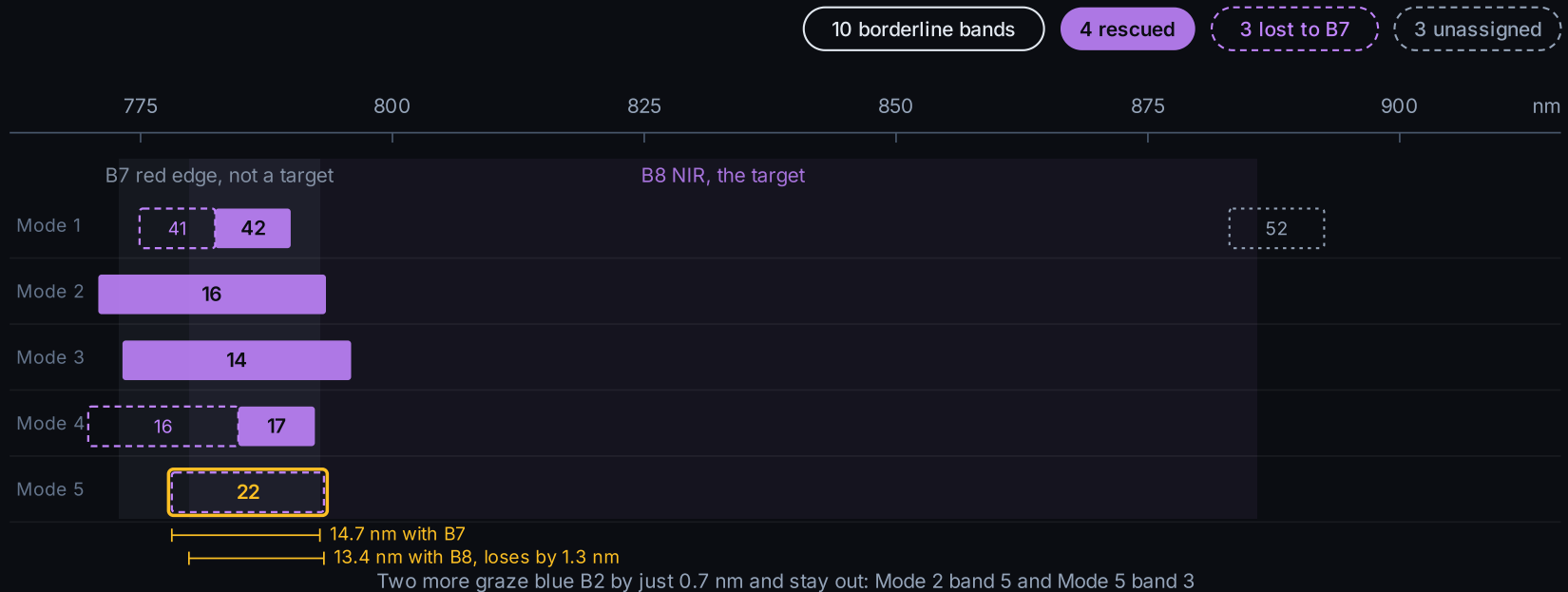
B, B2

NIR, B8

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	4-10 (7), 457-526 nm	13-15 (3), 547-578	23-25 (3), 647-678	42-51 (10), 782-883
2	3-4 (2), 485-517	6-7 (2), 555-580	10-12 (3), 644-686	16-17 (2), 771-887
3	2 (1), 486-497	4-5 (2), 546-577	7-8 (2), 656-683	14-15 (2), 773-890
4	1 (1), 484-496	2 (1), 545-558	4 (1), 664-675	17-18 (2), 785-800
5	2 (1), 485-496	4-5 (2), 545-576	7-8 (2), 654-681	23-26 (4), 793-896
20	1 (1), 469-500	2 (1), 500-593	3 (1), 609-689	4 (1), 782-901

Training mapping. Numbers are 1-based CHRIS band indices; nm is the spectral span. Stack order: R, G, B, NIR.

What the max overlap rule leaves out



Each of these bands touches the NIR target but the winner takes all rule sends it away. Four were rescued by hand for training, three were left out after losing the red edge B7 duel (including Mode 5 band 22, the heaviest NIR band the SRF sees), and three stay unassigned under the 30 percent overlap threshold.

METHOD 2 OF 3

SRF-weighted resampling

which every CHRIS band with nonzero S2A response takes part, no winner takes all

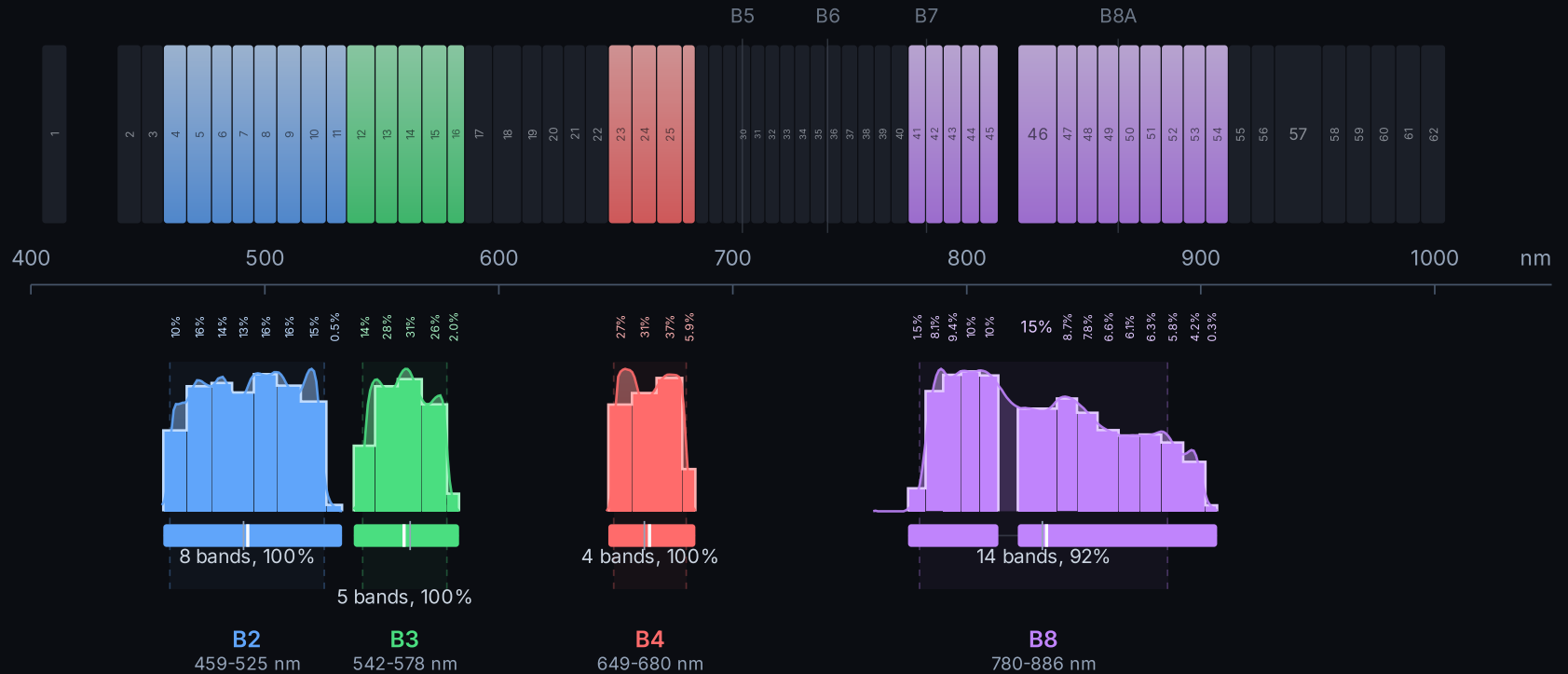
how w_i is the integral of the S2A response curve over the CHRIS band, and the pseudo-band is the weighted mean

then weights are normalized by the area CHRIS actually covers, not by the full SRF area

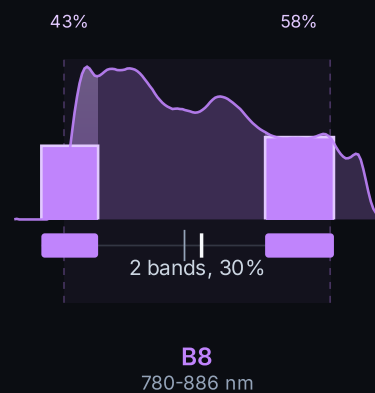
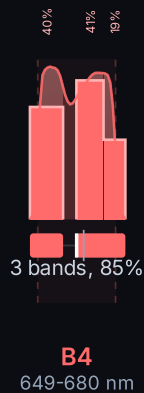
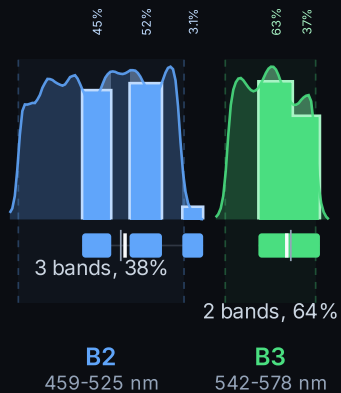
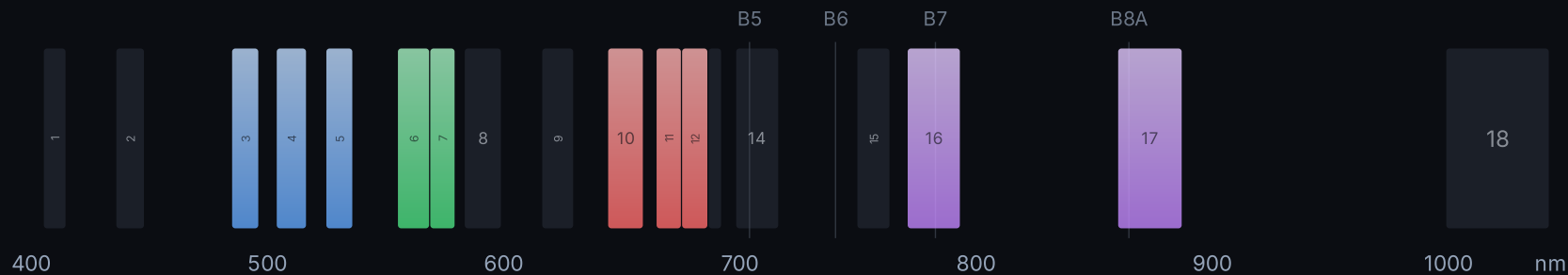
$$\tilde{L}_g = \frac{\sum_i w_i L_i}{\sum_i w_i}, \quad w_i = \int_{\lambda_i^{lo}}^{\lambda_i^{hi}} R_g(\lambda) d\lambda$$

Same four targets, read through the real S2A response curves instead of flat FWHM boxes. In the lanes: bands entering and the share of the response they cover; the white tick is the weighted centroid, the gray one the plain mean. The percentages above the lanes are each band's share of the mix.

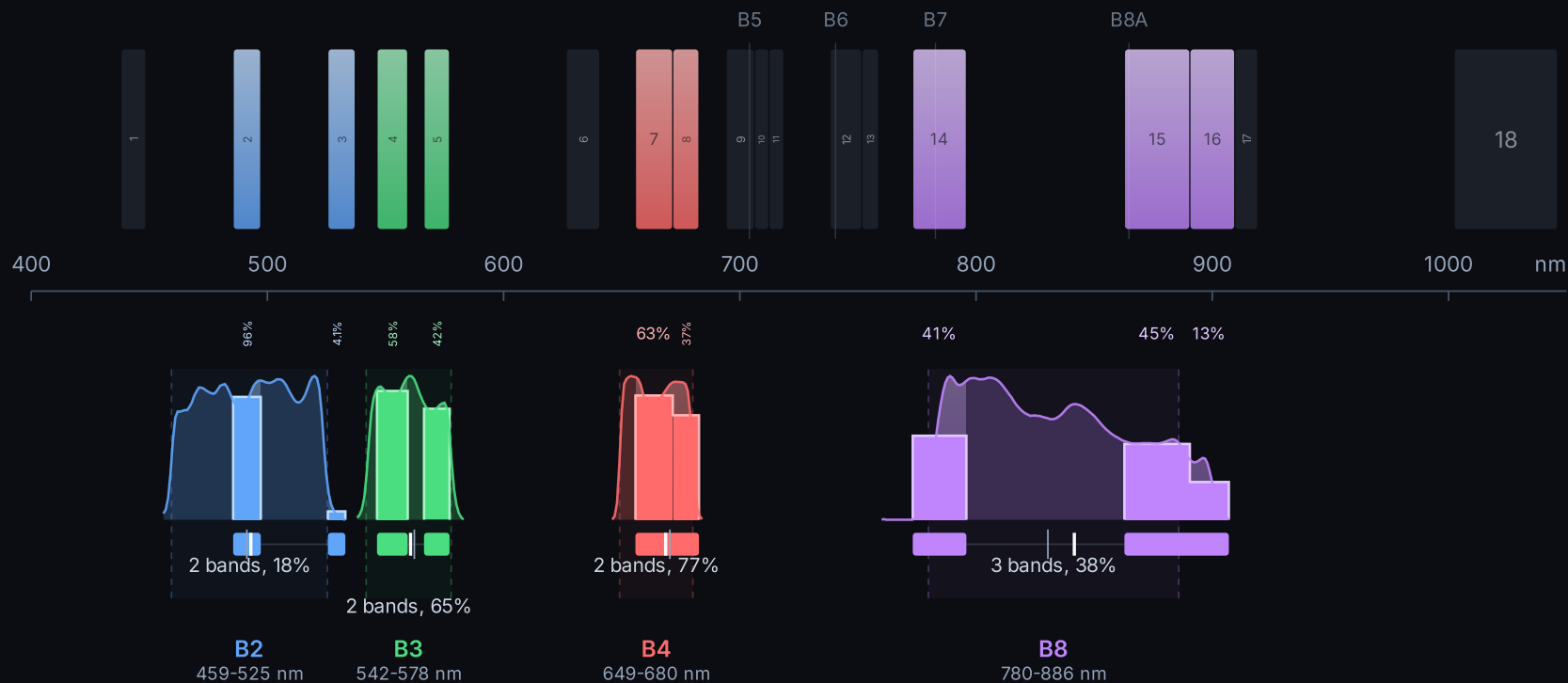
Mode 1 : 62 bands, ~34 m



Mode 2 : 18 bands, ~17 m



Mode 3 : 18 bands, ~13 m



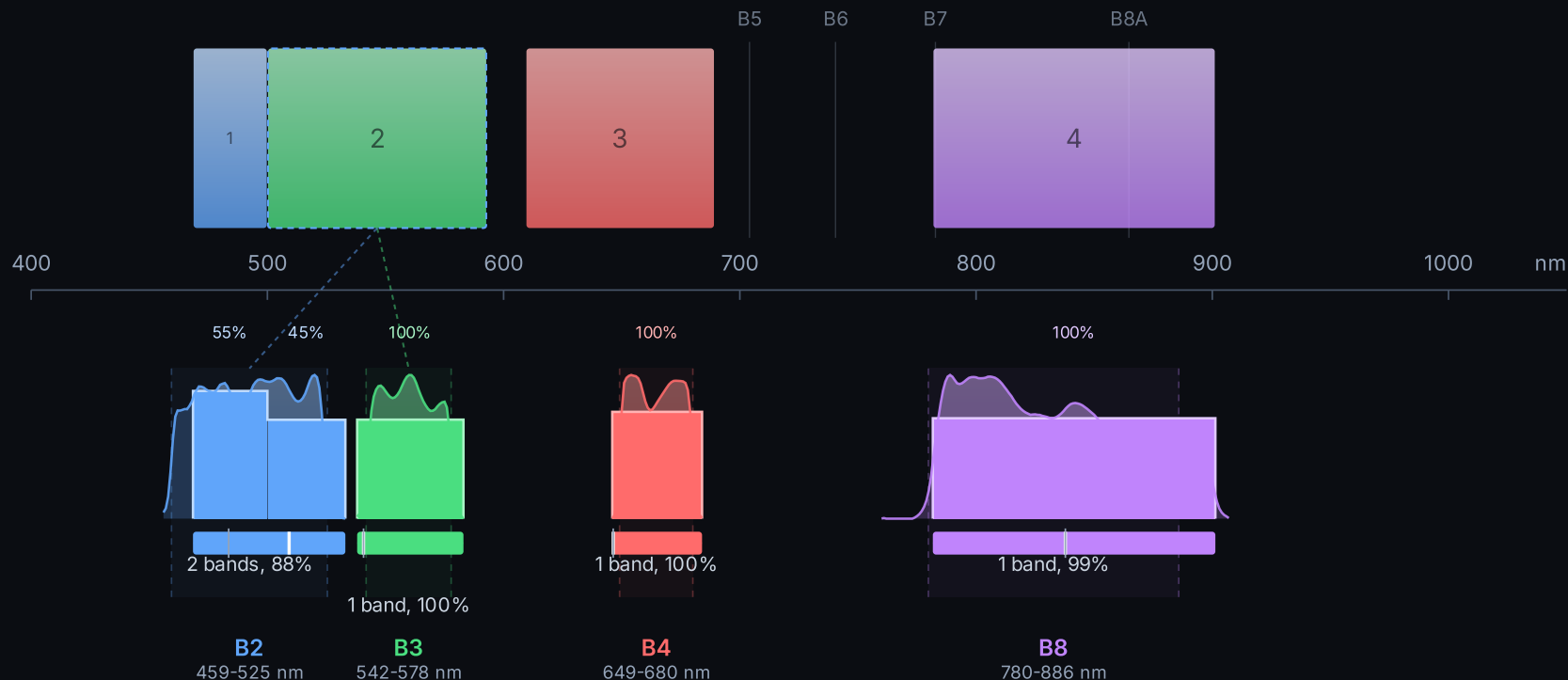
Mode 4 : 18 bands, ~11 m



Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Band 2 is 93 nm wide: it takes 45% of the blue mix next to band 1 and all of the green. That double duty is what pulls the blue centroid about 26 nm toward the green.

Plain mean vs SRF weights

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	7 → 8	3 → 5	3 → 4	10 → 14
2	2 → 3	2 → 2	3 → 3	2 → 2
3	1 → 2	2 → 2	2 → 2	2 → 3
4	1 → 1	1 → 1	1 → 2	2 → 3
5	1 → 2	2 → 2	2 → 3	4 → 7
20	1 → 2	1 → 1	1 → 1	1 → 1

CHRIS bands entering each pseudo-band. Left number is the plain mean set, right number the SRF set. Mode 5 band 22 comes back on its own, about a fifth of the B8 mix, and in mode 20 the SRF hands band 2 a share of the blue.

METHOD 3 OF 3

Convolution with the CHRIS response

which every CHRIS band whose modeled response touches the S2A curve takes part

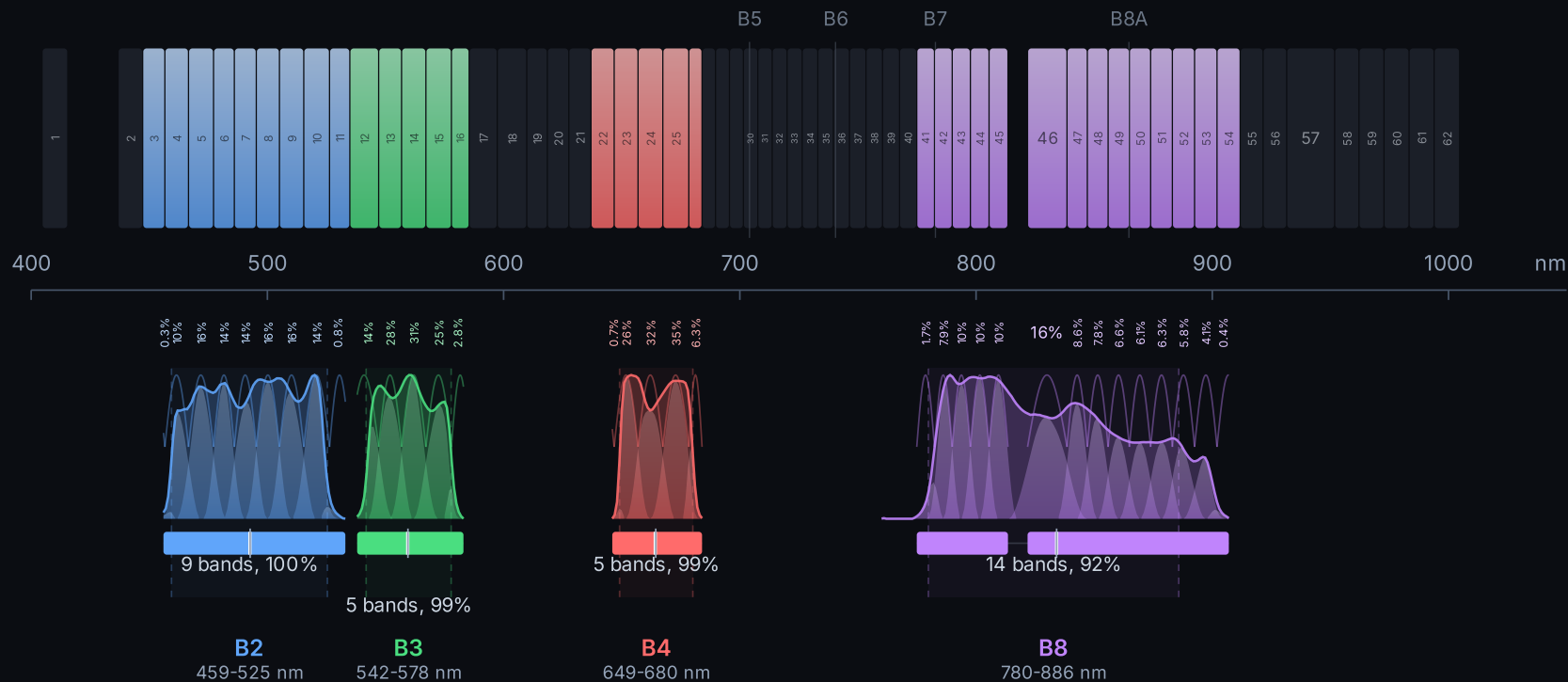
how w_i is the integral of the product of the two response curves

then CHRIS has no measured response, so R_i is a bell fitted to WILow, WIMid and WIHigh, with a Gaussian of FWHM = BWidth as the control

$$\hat{L}_g = \frac{\sum_i w_i L_i}{\sum_i w_i}, \quad w_i = \int R_g(\lambda) R_i(\lambda) d\lambda$$

The CHRIS response is modeled, not measured: no CHRIS SRF exists in the data, so the kernels are fitted to WILow, WIMid and WIHigh. In the lanes: the bell outline is the modeled response and the solid area is its product with the S2A curve, the band weight. The percentages above the lanes are each band's share of the group weight.

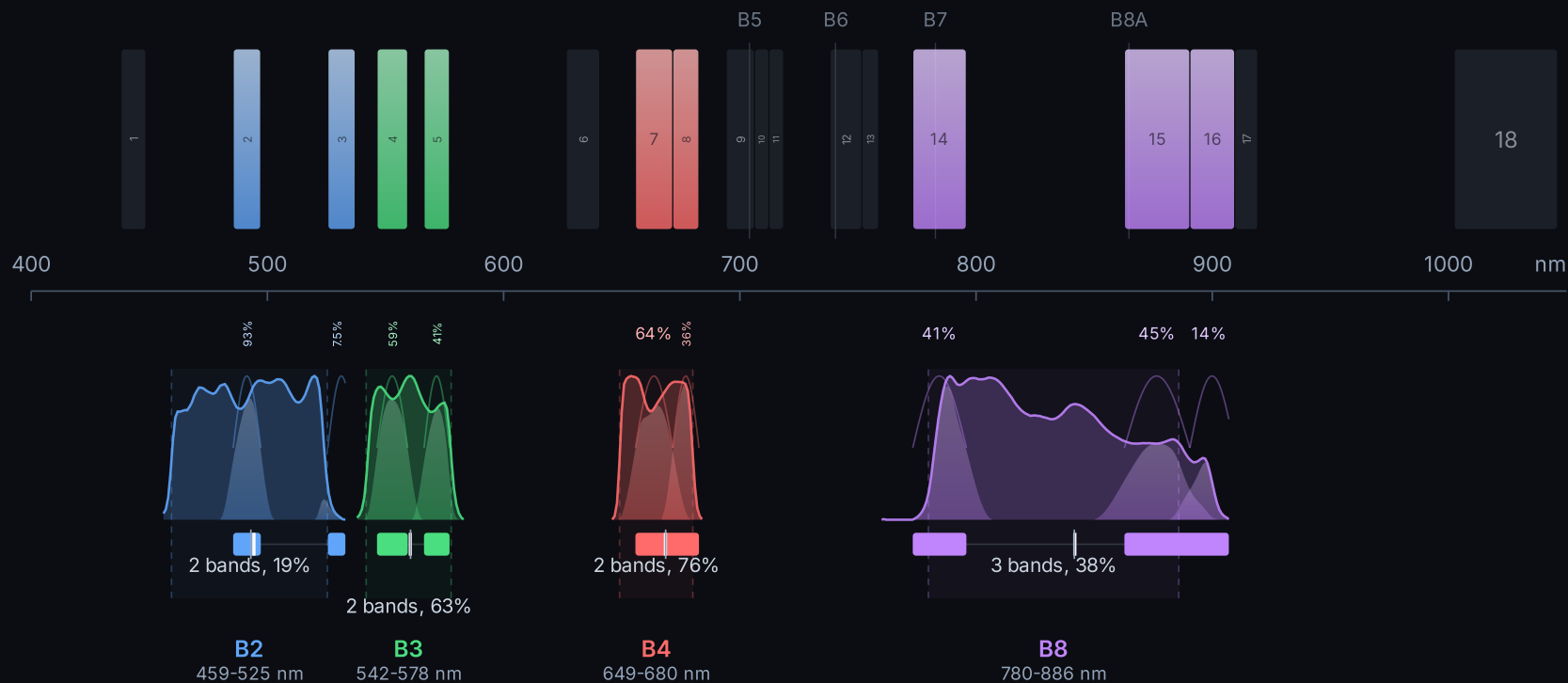
Mode 1 : 62 bands, ~34 m



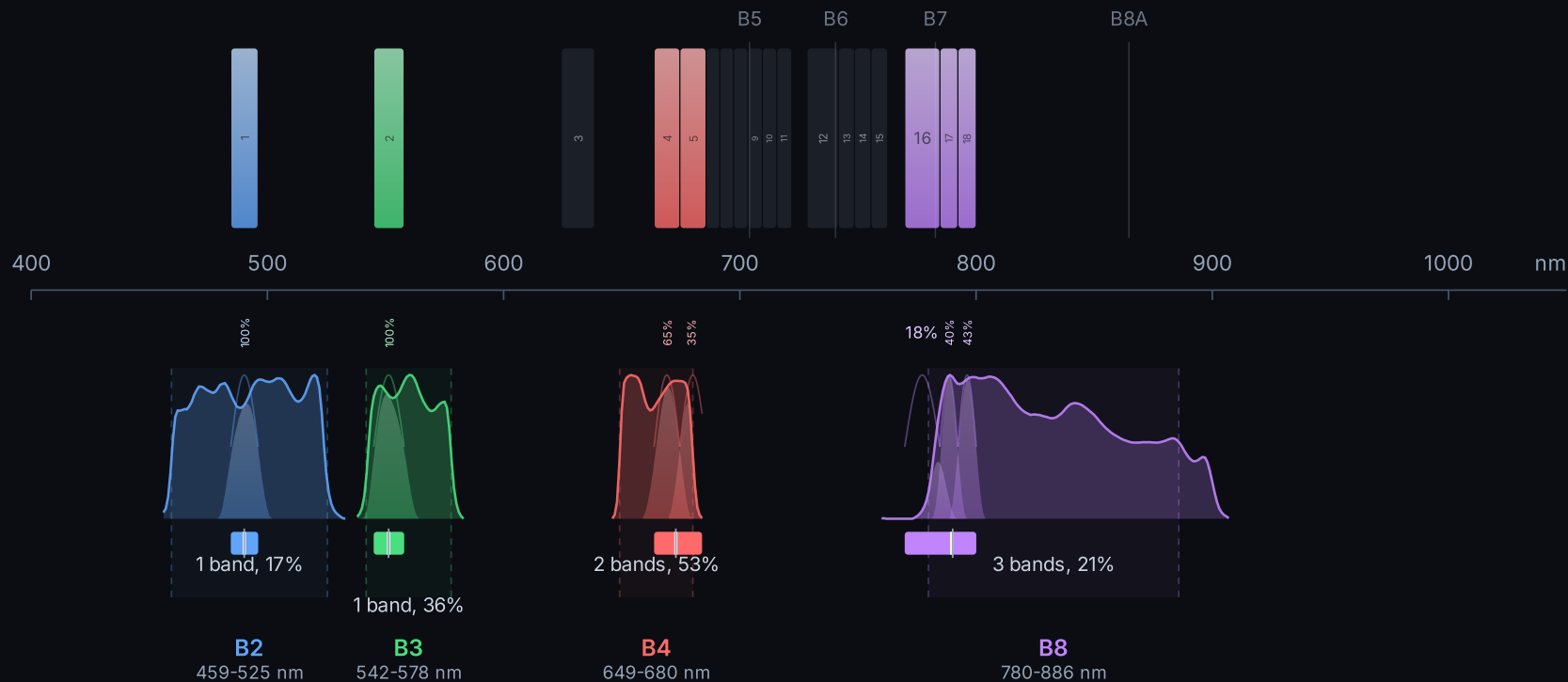
Mode 2 : 18 bands, ~17 m



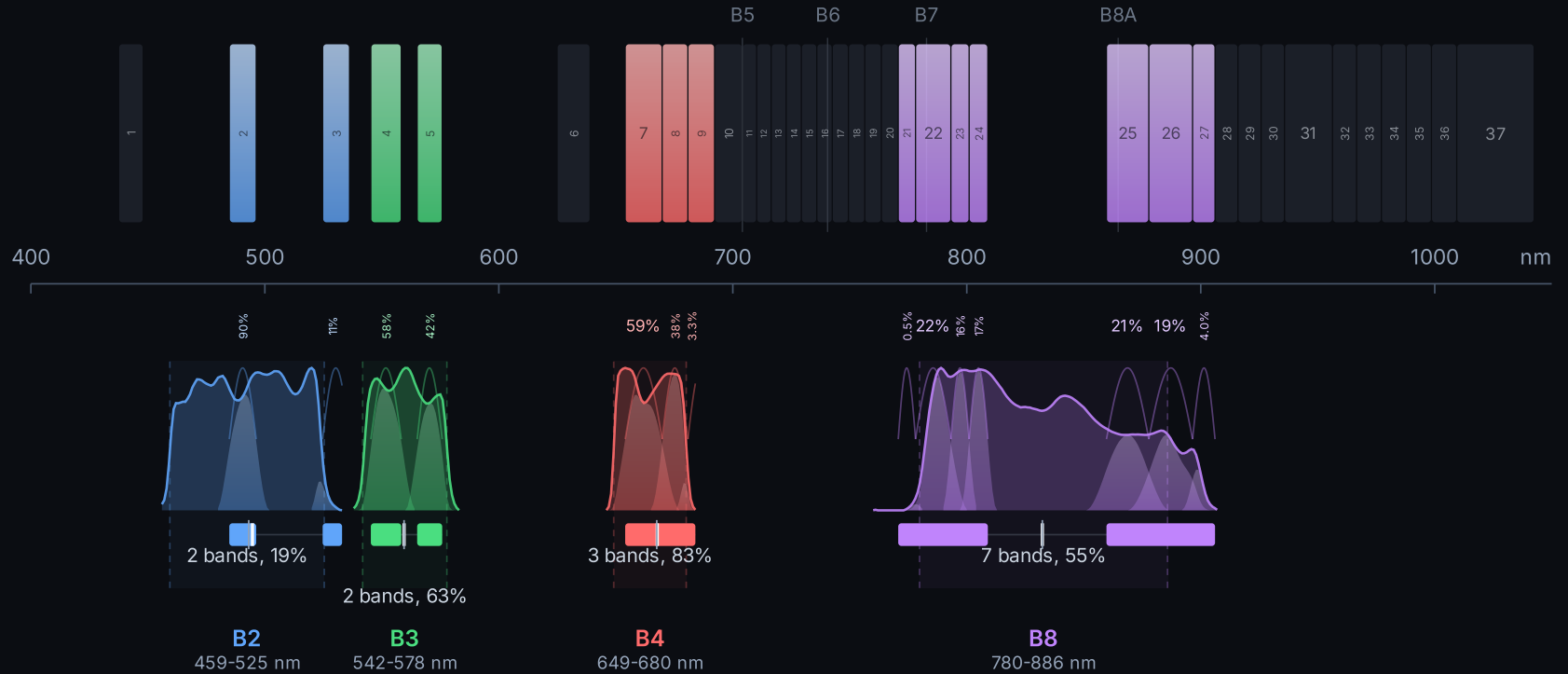
Mode 3 : 18 bands, ~13 m



Mode 4 : 18 bands, ~11 m



Mode 5 : 37 bands, ~20 m



Mode 20 : 4 bands, ~17 m, DN only



Bell vs Gaussian

Metric	Bell	Gaussian
Maximum modeled coverage	0.998 (Mode 1 B2)	1.061 (Mode 1 B2)
Bands shared across more than one target	1	5
Largest centroid shift against method 2 (nm)	1.6 (Mode 5 B2)	5.5 (Mode 20 B3)
Median RMSE of the emulated curve	0.361	0.353

Three-way comparison: band centroids

Mode	B2 blue	B3 green	B4 red	B8 NIR
1	490.9 → 492.7 → 492.6	562.1 → 559.5 → 559.4	662.2 → 664.4 → 664.2	832.3 → 834.1 → 834.1
2	500.2 → 501.9 → 502.5	567.9 → 566.3 → 566.3	667.4 → 664.6 → 664.7	827.8 → 834.6 → 834.5
3	491.3 → 493.0 → 494.3	562.2 → 560.6 → 560.5	670.4 → 668.6 → 668.5	830.4 → 841.6 → 841.8
4	490.2 → 490.2 → 490.2	551.3 → 551.3 → 551.3	669.1 → 672.9 → 672.9	792.4 → 790.1 → 789.7
5	490.5 → 493.1 → 494.7	561.0 → 559.5 → 559.5	668.5 → 667.5 → 667.7	839.5 → 832.3 → 832.3
20	483.6 → 509.2 → 509.5	540.7 → 540.7 → 540.7	646.4 → 646.4 → 646.4	837.8 → 837.8 → 837.8

Cells read plain mean → SRF weights → convolution, in nm of centroid. The kernel barely moves the centroid except where coverage is low (Mode 3 and Mode 5 B2) and in Mode 20.